

BAKANAS, Kazakhstan

WMO: 368210

Lat: 44.8117N

Lon: 76.2539E

Elev: 396

StdP: 96.66

Time Zone: 6.00 (E06)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.5	(c) -24.2	(d) -30.3	(e) 0.2	(f) -27.4	(g) -27.0	(h) 0.3	(i) -24.0	(j) 5.8	(k) -2.5	(l) 5.0	(m) -3.6	(n) 0.5	(o) 20	(p) 0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.5	(c) 37.2	(d) 19.6	(e) 35.5	(f) 18.8	(g) 33.9	(h) 18.4	(i) 21.3	(j) 32.3	(k) 20.2	(l) 31.4	(m) 19.5	(n) 30.5	(o) 1.9	(p) 340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.7	24.5	16.9	12.7	23.8	15.8	11.8	23.6	63.8	31.8	60.0	31.1	57.4	30.6	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.5	5.3	4.4	-29.2	39.9	4.1	1.6	-32.1	41.1	-34.5	42.0	-36.8	42.9	-39.7	44.1
			-29.4	22.6	4.0	1.9	-32.3	23.9	-34.6	25.0	-36.9	26.1	-39.8	27.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-9.7	-6.2	4.6	12.5	18.2	23.8	25.4	23.6	17.6	10.1	2.0	-6.8		
	DBStd	13.26	7.80	7.36	5.79	4.90	4.02	3.25	3.04	3.25	4.48	4.94	5.99	6.81		
	HDD10.0	2108	611	453	183	31	1	0	0	0	2	60	247	521		
	HDD18.3	3817	869	686	426	181	53	3	1	2	67	259	491	779		
	CDD10.0	1986	0	0	15	105	253	414	478	422	229	62	7	0		
	CDD18.3	653	0	0	0	5	48	167	219	166	44	3	1	0		
	CDH23.3	8884	0	0	4	156	751	2181	2899	2171	663	57	2	0		
	CDH26.7	4370	0	0	0	41	279	1098	1569	1105	266	12	0	0		
(14) Wind	WSAvg	1.4	1.1	1.2	1.7	1.7	1.8	1.5	1.5	1.4	1.2	1.2	1.2	1.1		
(15) Precipitation	PrecAvg	199	15	14	16	16	22	19	20	10	7	20	22	16		
	PrecMax	316	49	38	43	43	52	58	46	46	47	54	64	31		
	PrecMin	104	2	3	6	0	3	1	3	1	0	1	3	4		
	PrecStd	55	11	9	10	10	11	16	12	11	9	16	15	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	10.5	23.5	30.6	34.7	37.8	40.0	38.7	35.1	28.5	20.5	8.9		
		MCWB	3.5	5.8	12.5	16.4	18.2	19.9	20.3	20.4	18.0	15.4	11.7	4.7		
	2%	DB	3.8	6.9	19.4	27.3	32.0	35.9	37.7	36.5	32.3	24.7	16.0	5.2		
		MCWB	1.8	3.3	10.8	14.6	17.0	18.9	20.0	19.2	16.9	13.5	10.0	2.8		
	5%	DB	1.9	4.9	16.7	24.8	29.8	34.2	35.7	34.6	29.8	21.6	12.7	3.4		
		MCWB	0.4	2.3	9.6	13.7	16.2	18.4	19.1	18.5	15.7	11.9	8.2	1.6		
	10%	DB	0.3	2.9	13.9	22.0	27.7	32.5	33.9	32.5	27.3	18.9	9.8	1.7		
		MCWB	-0.8	1.0	8.5	12.6	15.4	18.0	18.6	18.0	14.8	10.8	6.5	0.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	6.5	13.3	17.5	19.5	21.9	23.2	23.0	19.0	15.9	12.6	5.2		
		MCDB	5.9	9.3	21.0	26.6	30.5	32.6	34.7	34.4	33.0	26.6	18.7	7.8		
	2%	WB	2.2	3.7	11.5	15.8	18.1	20.5	21.4	20.9	17.6	14.1	10.2	3.2		
		MCDB	3.6	6.3	18.2	24.8	28.8	31.4	32.4	32.3	29.9	22.3	14.8	5.0		
	5%	WB	0.6	2.4	10.1	14.5	17.2	19.6	20.4	19.6	16.5	12.9	8.5	1.7		
		MCDB	1.7	4.6	15.8	22.6	27.0	30.5	31.6	31.1	27.9	20.1	12.3	3.2		
	10%	WB	-0.7	1.1	8.6	13.3	16.2	18.9	19.6	18.7	15.4	11.4	6.8	0.3		
		MCDB	0.3	2.8	13.8	20.6	25.5	29.5	30.6	30.1	26.0	17.9	9.6	1.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	9.9	11.3	13.9	15.2	14.7	14.5	15.6	15.6	13.3	9.6	8.5		
		MCDBR	9.8	10.5	15.9	18.8	18.2	17.2	17.5	18.3	19.0	17.6	13.9	9.7		
	5% WB	MCWBR	7.9	7.3	8.4	8.8	7.3	5.6	5.7	6.8	8.1	8.8	8.3	7.4		
		MCWBR	8.5	9.8	14.5	16.2	15.4	14.4	14.4	16.0	17.1	15.3	12.6	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.340	0.380	0.421	0.453	0.446	0.445	0.442	0.405	0.380	0.394	0.376	0.339		
	taud		2.089	2.034	2.018	1.984	2.058	2.121	2.145	2.231	2.248	2.121	2.106	2.113		
	Ebn at Noon		731	771	796	807	827	828	825	845	834	748	681	686		
	Edh at Noon		100	129	150	169	162	153	147	130	118	116	99	89		
(44) All-Sky Solar Radiation	RadAvg		1.68	2.67	4.01	5.53	6.95	7.31	7.09	6.52	5.11	3.22	1.87	1.28		
	RadStd		0.19	0.23	0.27	0.37	0.31	0.31	0.41	0.27	0.20	0.25	0.14	0.13		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-1.18	-1.67	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	-0.77	-1.20	-203	-252	+139	+72

Nomenclature: See separate page